1) The geometric constraint is:

$h(q) = x^2 + y^2 + z^2 - r = 0$. The corresponding kin. constraint is:

$$\dot{h} = \frac{\partial h}{\partial q}\dot{q} = 2x\dot{x} + 2y\dot{y} + 2z\dot{z} = A^T\dot{q} = 0$$

Of course is not controllable, so global mobility isn't achievable due to the presence of the geometric constr. In Fact, the DoF are 2, and in each conf. the admissible velocities lies on the plane tangent to the sphere. We Find a basis for $\mathcal{N}\left((2x \quad 2y \quad 2z)^T\right)$:

$$g_1 = (0 \quad -2z \quad -2y)^T$$
$$g_2 = (-2y \quad -2x \quad 0)^T$$
$$\Rightarrow \begin{cases} \dot{x} = -2y\,u_2 \\ \dot{y} = -2z\,u_1 - 2x\,u_2 \\ \dot{z} = -2y\,u_1 \end{cases}$$

Consider $[g_1, g_2] = \frac{\partial g_2}{\partial q} g_1 - \frac{\partial g_1}{\partial q} g_2 =$

$$\begin{pmatrix} 0 & -2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ -2z \\ -2y \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -2 \\ 0 & -2 & 0 \end{pmatrix}\begin{pmatrix} 2y \\ 2x \\ 0 \end{pmatrix} = \begin{pmatrix} 4z \\ 0 \\ -4x \end{pmatrix}$$

is not linearly independent From g_1 and g_2, hence the model is not controllable.

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2) A (2,3) chained form system is:

$$\begin{cases} \dot{z}_1 = v_1 \\ \dot{z}_2 = v_2 \\ \dot{z}_3 = z_2 v_1 \end{cases}$$

Consider any initial conf. z_i, and a Final conf. z_s.

- Phase 1:

$$\begin{cases} z_1(t) = z_{i1} + v_1 t \\ z_2(t) = z_{i2} + v_2 t \end{cases} \quad \text{apply} \quad \begin{matrix} v_1 = \frac{1}{T}(z_{s1} - z_{i1}) \\ v_2 = \frac{1}{T}(z_{s2} - z_{i2}) \end{matrix} \quad \text{in } t \in [0, T]$$

At the end $z_1(T) = z_{1s}$, $z_2(T) = z_{2s}$.

$$\dot{z}_3(t) = \frac{1}{T}(z_{s1} - z_{i1})(z_{i2} + v_2 t)$$

We denote $z_3(T) = \tilde{z}_3$.

- Phase 2:

i want to apply a displacement Δ s.t. $\Delta + \tilde{z}_3 = z_{3s} \Rightarrow \Delta = z_{3s} - \tilde{z}_3$

We apply the periodic control inputs with $a = \omega = 1$, $b = \frac{z_{3s} - \tilde{z}_3}{\pi}$

so $ab\frac{\pi}{\omega^2} = b\pi = \frac{z_{3s} - \tilde{z}_3}{\pi}\pi = z_{3s} - \tilde{z}_3$ ✓

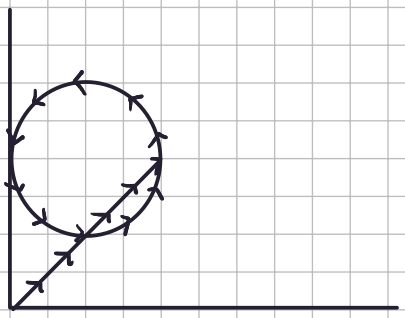
I choose arbitrarely T, after choosing a, ω, b becomes constrained.

For the example with $z_0 = 0$ $z_s = (1\ 1\ 1)^T$, let $T = 1$

$$\dot{z}_3(t) = \frac{1}{T}(z_{s1} - z_{i1})\left(z_{i2} + \frac{1}{T}(z_{s2} - z_{i2})\,t\right) = t$$

since $z_3(0) = 0 \Rightarrow z_3(T) = \int_0^T t\,dt = \frac{t^2}{2}\Big|_{t=T=1} = \frac{1}{2} = \tilde{z}_3$

So we want $\Delta = z_{3s} - \tilde{z}_3 = \frac{1}{2}$. For $a = b = 1$:



$$ab\,\frac{\pi}{\omega^2} = \Delta \Rightarrow \frac{\pi}{\omega^2} = \frac{1}{2} \Rightarrow \omega^2 = \frac{2}{\pi} \Rightarrow \omega = \sqrt{2/\pi}$$ so the total time is $1 + \frac{1}{\sqrt{2/\pi}}2\pi$

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3) This is a system with 4 inputs and 2 coordinates.

Robot 1:
$$\begin{cases} \dot{x}_1 = u_1 \cos\theta_1 \\ \dot{\zeta}_1 = u_1 \sin\theta_1 \\ \dot{\theta}_1 = u_2 \end{cases}$$

Robot 2:
$$\begin{cases} \dot{x}_2 = u_3 \cos\theta_2 \\ \dot{\zeta}_2 = u_3 \sin\theta_2 \\ \dot{\theta}_2 = u_4 \end{cases}$$

$\Rightarrow$

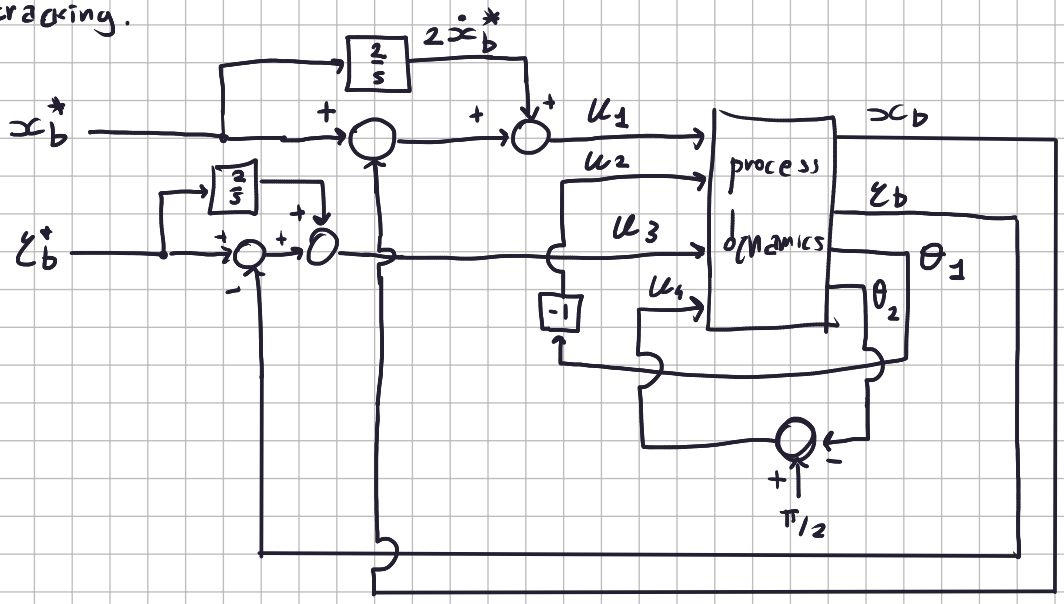
$$\begin{cases} \dot{x}_b = \frac{1}{2}(u_1\cos\theta_1 + u_3\cos\theta_2) \\ \dot{\zeta}_b = \frac{1}{2}(u_1\sin\theta_1 + u_3\sin\theta_2) \\ \dot{\theta}_1 = u_2 \\ \dot{\theta}_2 = u_4 \end{cases}$$

I consider:

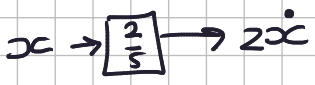
$u_2 = -\theta_1$ $\quad u_4 = \frac{\pi}{2} - \theta_2$ hence $\lim_{t\to\infty}(\theta_1, \theta_2) = (0\ \ \frac{\pi}{2})$

hence $\lim_{t\to\infty}(\dot{x}_b, \dot{\zeta}_b) = (\frac{1}{2}u_1, \frac{1}{2}u_3)$, so by setting

$u_1 = x_b^* - x_b + 2\dot{x}_b^*$, $u_3 = \zeta_b^* - \zeta_b + 2\dot{\zeta}_b^*$ (Feedback + FF) we achieve the desired tracking.



note



we must know $x_1, x_2, \zeta_1, \zeta_2$ θ_1, θ_2.

To estimate all the variables, we apply the EKF for both the two robots.

At time step k, our conf. have the distribution $\mathcal{N}(\hat{q}_k, P_k)$

$\hat{q}_k = (\hat{x}_{1k} \;\; \hat{y}_{1k} \;\; \hat{\theta}_{1k} \;\; \hat{x}_{2k} \;\; \hat{y}_{2k} \;\; \hat{\theta}_{2k})^T$

$u_k = (u_{1k} \;\; u_{2k} \;\; u_{3k} \;\; u_{4k})^T$ We consider $\tilde{q}_{k+1|k} = f(\hat{q}_k)$ as a first estimate:

$$f(\hat{q}_k) = \begin{cases} \hat{x}_{1k} + T_s u_{1k} \cos(\hat{\theta}_{1k}) \\ \hat{y}_{1k} + T_s u_{1k} \sin(\hat{\theta}_{1k}) \\ \hat{\theta}_{1k} + T_s u_{2k} \\ \hat{x}_{2k} + T_s u_{3k} \cos(\hat{\theta}_{2k}) \\ \hat{y}_{2k} + T_s u_{3k} \sin(\hat{\theta}_{2k}) \\ \hat{\theta}_{2k} + T_s u_{4k} \end{cases}$$

let $F_k = \left.\frac{\partial f}{\partial q}\right|_{q=\hat{q}_k, u=u_k}$ and $P_{k+1|k} = F_k P_k F_k^T + V_k$

we are assuming a noise v_k on the dynamics: $v_k \sim \mathcal{N}(0, V_k)$

$$F_k = \begin{bmatrix} 1 & 0 & -T_s u_{1k} \sin\hat{\theta}_{1k} & & & \\ 0 & 1 & T_s u_{1k} \cos\hat{\theta}_{1k} & & & \\ 0 & 0 & 1 & & & \\ & & & 1 & 0 & -T_s u_{3k} \sin\hat{\theta}_{2k} \\ & & & 0 & 1 & T_s u_{3k} \cos\hat{\theta}_{2k} \\ & & & 0 & 0 & 1 \end{bmatrix}$$

Now we have to define the measurement output.

Let $(x_L \;\; y_L)^T$ be the position of the landmark. We have 3 outputs:

$$z_k = h(q_k) = \begin{cases} ((x_{1k} - x_L)^2 + (y_{1k} - y_L)^2)^{1/2} \\ ((x_{2k} - x_L)^2 + (y_{2k} - y_L)^2)^{1/2} \\ ((x_{1k} - x_{2k})^2 + (y_{1k} - y_{2k})^2)^{1/2} \end{cases} \qquad H_{k+1} = \left.\frac{\partial h}{\partial q}\right|_{q=\tilde{q}_{k+1}}$$

z_{k+1} is the measured output, the innovation is defined as $v_{k+1} = z_{k+1} - h(\tilde{q}_{k+1|k})$ and is used to correct $\hat{q}_{k+1|k}$ to $\hat{q}_{k+1}$ through the Kalman gain matrix. Then $P_{k+1|k}$ is corrected with H_{k+1}.

